Suggested Materials:

Raspberry Pi Zero W –

https://www.adafruit.com/product/3400

SHT31 Humidity Sensor Board or equivalent (i2c recommended)- https://www.adafruit.com/product/2857 (purchasable on alternate retailers. Alternatives can be used but the included sample code will need to be altered to match the new sensor)

Soldering Iron-

https://a.co/d/04slcJeu (nothing fancy, though a nicer soldering iron is an investment if multiple projects are being pursued)

Solder-

https://a.co/d/01JolSWb (not necessary if soldering iron kit includes solder, but more will be needed if multiple projects are being pursued)

Wire –

https://a.co/d/00qnlKwV (multiple colors of wire has proven helpful in keeping organized, especially on larger projects

Micro SD Card

(any microSD card can be used, it does not have to be large)

Micro USB Power Cable

https://www.adafruit.com/product/1995

Micro USB to USB converter

(any converter will do)

HDMI to Mini HDMI Cable

(any cable will do)

Setup-

Follow all manufacturer guidelines for setting up your control board. For this project, I have used a Raspberry Pi Zero W, because I have found that ecosystem to be the easiest to work with. The primary advantage in my eyes is the traditional desktop display, allowing me to

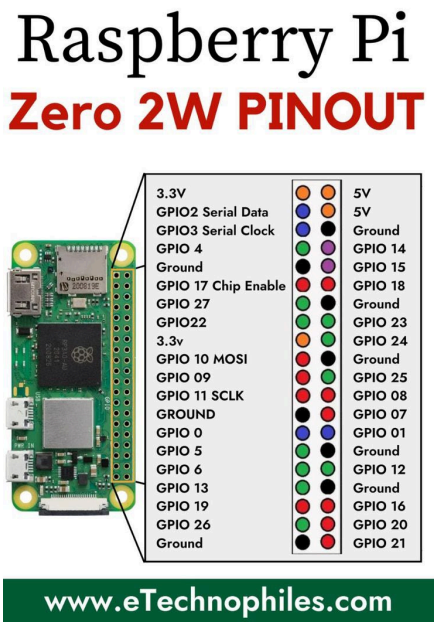
write the code on the computer itself, rather than using a remote connection from a different device. Once the device is formatted you will need to connect it to a wifi network in the location it will live. If the device is ever moved, or the wifi network changes, simply plug it into a monitor and get it connected to the new one. You will need to enable i2c on your board, this guide will show you two simple ways of doing so https://www.youtube.com/watch?v=xaTySaEXvSE&t=1s

IoT Platform

For this project I used AdafruitIO as the platform to run the data through. This allows me to access the logger information anywhere I have internet access. AdafruitIO is intuitive and easy to use, with the added benefit of having a basic free tier. The data visualizations and block-based builders are also notably user friendly. After creating an account, you will need to make two feeds, “temperature” and “humidity” (pay attention to the capitalizations you use for the names, they need to match your code).

Environmental Sensor

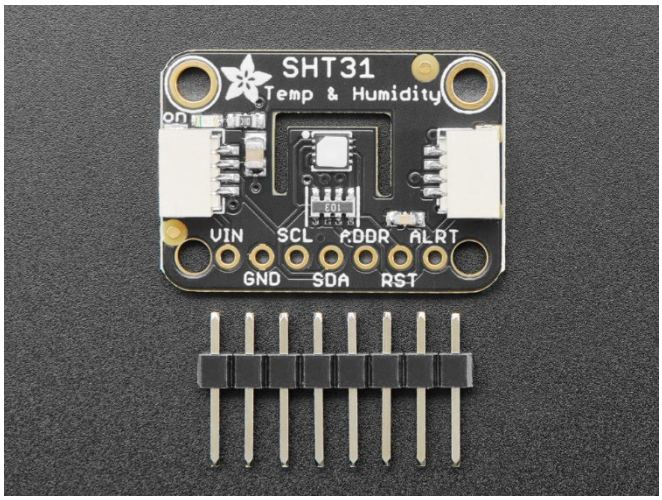
I initially chose the Adafruit SHT31 sensor because it was inexpensive and had a greater degree of RH% accuracy compared to other sensors. This sensor is no longer available from Adafruit, but it looks like it can be purchased from their distribution partner at the time of writing this (3/26). This sensor does not need to be used for this project, but changing sensor will require alterations to the provided project code. Always read the documentation for the sensor you are using, as they will have their own guide for using it in your coding projects. If a different sensor is used, I would recommend one that is i2c compatible, because it makes the wiring and coding much easier.



Wiring the Sensor

There are multiple ways to wire sensors to boards. In my project I first soldered pins to the board itself and used socket wires to create a plug, allowing the removal of the sensor. Were I to do this again, I would solder the wires directly to both boards, so this is what I would recommend here. There are a number of tutorials on the

internet about soldering, so I will leave that to you. The image to the right shows the purpose of each pin socket for the board. The important pin sockets for our project are GPIO 2 Serial Data, GPIO 3 Serial Clock, 3.3V and Ground. Using these four pin sockets we will power the sensor with the board and allow the transfer of data using i2c. The sensor board has 7 pin sockets, but we will only be using 4. VIN, GND, SCL, and SDA. VIN will be connected to the 3.3v. GND will be wired to Ground, SCL will be connected to GPIO Serial Clock, and SDA will be wired to GPIO 2 Serial Data. If you are using a different sensor the connections will all, be the same, assuming you are using an i2c compatible sensor. With this, your sensor is now connected to the board itself.



Using the Sample Code

Using the sample code provided you will be able to create your own AdafruitIO feeds and dashboards to display collected data. You will need to make some slight changes to the sample code however. You will need to enter your AdafruitIO Active Key and username in the code, so the program knows where to send the data. Once this code is updated to reflect your own information, it should be good. This program is intended to run once, measure the humidity and temperature, upload that data to AdafruitIO, and then close the program. That means it will only ever give one data point for each criteria. How do we solve this?

> Look for the following lines in the code and replace the stars with your own Adafruit IO information.
>
> ADAFRUIT_IO_KEY = '**********'
>
> ADAFRUIT_IO_USERNAME = '**********'

Scheduling the Script

If the sample code program is only supposed to run once and then stop, we have to find a method of automatically running it at certain intervals. This is where the Crontab tool comes in handy. Using Crontab you can schedule parameters for the script to automatically be run. Before going in to this it's a good idea to try and run your script once

to make sure the hardware and software components are working together, and AdafruitIO is receiving your data. Additionally, it is a good idea to write down or otherwise note your full file path for the script, as you will need that to tell the computer to run it. Finally, you should have an idea for when you want this program to gather data. In my project I have it gathering data hourly, but you can choose frequencies tailored to your needs. You will need to open the command prompt on your pie to get started. I will include a video tutorial that gives an overview of the crontab process and how you can use it to schedule your script.

https://www.youtube.com/watch?v=4xis3DNGNIY&t=203s

Conclusion

If all has gone right, you should have a working data logger now. The process was overwhelming initially, but after exposure and a lot of googling, I was able to get this working with no prior knowledge of python, Raspberry Pi computers, or soldering. Everything in this project was learned through googling and scrolling forums. There are numerous resources available and there is never a problem you'll encounter that someone else hasn't encountered before. I will include links to guides and posts that were helpful in the course of this project.

https://learn.adafruit.com/adafruit-io-basics-temperature-and-humidity/overview

This guide covers a very similar project to ours, but notably uses a different humidity sensor. This guide is useful for showing you the overview AdafruitIO and implementing it within a python script for something similar to our project.

https://learn.adafruit.com/adafruit-sht31-d-temperature-and-humidity-sensor-breakout/overview

This is less of a guide, and more a documentation of the SHT31-D sensor that I have used in this project. The most important section is the Python installation guide. This will instruct you on the process to download the necessary libraries to be able to use the sensor in a python script. **Please be aware that circuit python is different than python. We are using Python, not circuit python.**